



Review

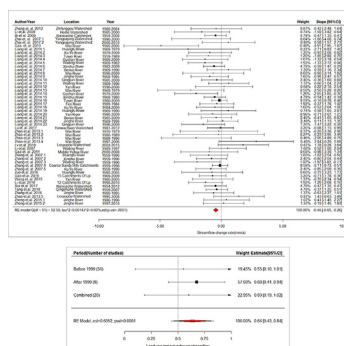
Impacts of land use change and climatic effects on streamflow in the Chinese Loess Plateau: A meta-analysis

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HIGHLIGHTS

- First meta-analysis to determine the land use and climate change impacts on the streamflow.
- Fluctuation base on hydrological model results in the Chinese Loess Plateau.
- 61 case studies were analyzed from 25 watersheds in 22 published literatures.
- Streamflow was estimated to reduce by -0.46 mmm/year.
- Land use change accounted for 63.52% of the streamflow reduction.
- Climate change accounted for 36.48% of the streamflow reduction.

GRAPHICAL ABSTRACT



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ABSTRACT

Land use and climate change are recognized as two major drivers affecting surface streamflow. On the Chinese Loess Plateau, implementation of several land restoration projects has changed land cover in recent decades. The main objectives of this study were to understand how streamflow evolved on the Loess Plateau and how land use and climate change have contributed to this change. In this study, we selected 22 hydrological modelling studies covering 25 different watersheds in the Loess Plateau and we performed a meta-analysis by using the hydrological and meteorological data collected from these studies. The results indicate a streamflow decrease in 41 of a total of 52 case studies whereas precipitation change was found to be non-significant in the majority of the cases. Streamflow reduction was estimated to be -0.46 mm/year by meta-analysis across all case studies. Land use change was estimated to have 63.52% impact on the streamflow reduction whereas climate change accounted for 36.48% of the impact. Using meta-regression, an increasing soil and water conservation area was found to be positively correlated to streamflow reduction. We conclude that in the Chinese Loess Plateau, streamflow shows a decreasing trend and land restoration is the major cause of this reduction. To the knowledge of the authors, this is the first study that estimates streamflow dynamics across many watersheds on the entire Loess Plateau.

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1. Introduction

According to observational evidence from most regions in the world, land cover and climate variability are the two main factors influencing hydrological flow (Huntington, 2006; Changnon and Demissie, 1996; Li et al., 2009). Labat et al. (2004) demonstrated that climate change is leading to a continental precipitation increase, which results in an intensification of the global hydrological cycle. Simultaneously, human activities have altered the spatial-temporal distribution of water resources which has contributed to fluctuations of surface hydrology (Milly et al., 2005). Extremes of surface streamflow may cause natural disasters such as floods and droughts, which is threatening human life and property. For the sake of safeguarding human society and to avoid economic loss as a results of floods and droughts, investigating the impacts of climate change and human activities on streamflow is becoming an important scientific issue (Zhao et al., 2014). Understanding how streamflow is affected by land cover and climate change is crucial to inform adaptive land and water management. Assessment of human activities and climate change impacts on streamflow is usually performed by applying various hydrological models at regional scale (Akter et al., 2018; de Hipt et al. 2019a, b). For instance, Zipper et al. (2018) developed a regression-based framework to separate climate and land use effects on hydrological fluxes in the Yahara River Watershed in Wisconsin, USA. The Soil Water Assessment Tool (SWAT) model is also widely used to understand the long-term impacts of watershed management and climate change (Ghaffari et al., 2010; Lin et al., 2015; Narsimlu et al., 2013).

To address land degradation and ecological deterioration issues, a number of large-scale land restoration projects have been implemented around the world which have significantly changed the local land use and land cover (LUCC) (Benayas et al., 2009). As an example of large-scale LUCC programs, China has initiated land restoration projects to recover large-scale degraded land since the 1950s. The Chinese Loess Plateau, as one of the most severely eroded regions in the world, was given strong attention in

government-led land restoration policies (Sun et al., 2014). From 1970 till the end of the last century, multiple land restoration projects focusing on vegetation rehabilitation, soil and water conservation measures (SWC) and desertification control were implemented in the Loess Plateau (Deng et al., 2019). In 1999, one of the world's largest ecological restoration programs named Grain to Green project (GTGP) was initiated with the aim to recover forest and grassland on formerly bare and cultivated sloping land. The GTGP is well-known for its large implementation area and ambitious goals (Zhou et al. 2016; Jian et al. 2015). GTGP has achieved initial success on vegetation recovery; for instance, from 1998 to 2005, forest and grassland cover in Shaanxi province raised from 29.7% to 42.2% (Cao et al. 2009).

Previous studies have investigated the hydrology of Loess Plateau watersheds at varying regional and catchment scales with various methods, most commonly using hydrological models and field/plot experiments (Zhao et al. 2017; Li et al. 2007). However, there has been little quantitative impact analysis reviewing previous findings systematically. Hu et al. (2017) concluded that ecological restoration accounted for 72.18% of the streamflow reduction by means of a meta-analysis based on field experimental data in the Loess Plateau. However, little attention has been paid to summarizing and synthesizing results from hydrological modelling studies. Systematic review, combined with meta-analysis are considered as good methods to evaluate and monitor the performance of ecological restoration (Crouzeilles et al. 2016). A meta-analysis is a statistical method to synthesize data from a series of studies, providing an objective, transparent and reliable framework (Borenstein et al. 2011; Akter et al. 2018). In this study, using a meta-analysis, we integrated results from hydrological modelling studies to quantify the effects of ecological restoration and climate change on streamflow. Moreover, different SWC measures have been applied in land restoration projects, and their effects have been assessed in various hydrological modelling studies (Chen et al. 2017; Dou et al. 2009). This study seeks to investigate the consistency of findings emerging from modelling and to indicate the key factors affecting streamflow.

The aims of this study therefore were to: a) Quantify the temporal trends in precipitation and surface streamflow; b) Identify the land use and climate impacts on streamflow; c) Reveal impact levels of different soil and water conservation measures on streamflow. This study is the first to analyse these changes for the entire Loess Plateau systematically by using meta-analysis, which is a powerful method that minimizes diversity of results caused by differences in methods used in the analysed studies.

2. Materials and methods

2.1. Study area

The Chinese Loess Plateau is located in central China in the upper and middle reaches of the Yellow River (Fig. 1; from 100°54' to 114°33'E and 33°43' to 41°16'N) and covers a total area of 624,000 km². The Plateau stretches across seven administrative areas including Qinghai, Gansu, Ningxia, Shaanxi, Shanxi, Inner Mongolia and Henan provinces. The average precipitation ranges between 200 and 600 mm and gradually decreases from southeast to northwest (Shi and Shao, 2000). The elevation of the Loess Plateau ranges from 4917 to 200 to m.a.s.l, with the altitude increasing from southeast to northwest (Fig. 1).

2.2. Article search and data extraction

To quantify land-use and climate impacts on surface water flow in the Chinese Loess Plateau, an article search was conducted using various search engines. English articles were mainly collected from

Web of Knowledge (<http://www.isiwebofknowledge.com/>), Science Direct (<https://www.sciencedirect.com/>) and Google Scholar (<https://scholar.google.com/>); meanwhile China National Knowledge Infrastructure (<http://www.cnki.net>) was used for searching articles written in the Chinese language. The article search was carried out up to 18th of May in 2018 with restricted publication year from 1945 to 2018, in which keywords were combined as : ("runoff" OR "surface flow" OR "discharge" OR "streamflow" OR "hydrological" OR "water yield") AND ("land use" OR "human" OR "restoration" OR "climatic" OR "climate") AND ("Chinese" OR "Loess Plateau" OR "Yellow River"). The article selection process is explained in Fig. 2. Web of Knowledge and Science Direct were the main sources of articles. Articles meeting the following conditions were considered: 1) both land use change and climate change influences are included and impacts on surface water flow are studied at the same time; 2) the research site was located within the Chinese Loess Plateau, i.e. the middle Yellow River basin is out of the Loess Plateau scope and thus excluded; 3) quantitative information on land use change, runoff and precipitation are directly provided or could be indirectly estimated from the article.

Overall, data extracted from the selected articles comprised land use data, meteorological data and impact factor data. Following Hu et al. (2017), Engauge Digitizer was used to extract numerical data from scatter plot and bar plot figures, yielding yearly precipitation and runoff data from the articles. The land use change data were collected from tables in the article including land use type, land use area, drainage location, year and total basin area. Where reported, impact factor data to identify the impacts from land use change and climate change were collected from the selected articles, and were usually presented in tables. Calculation

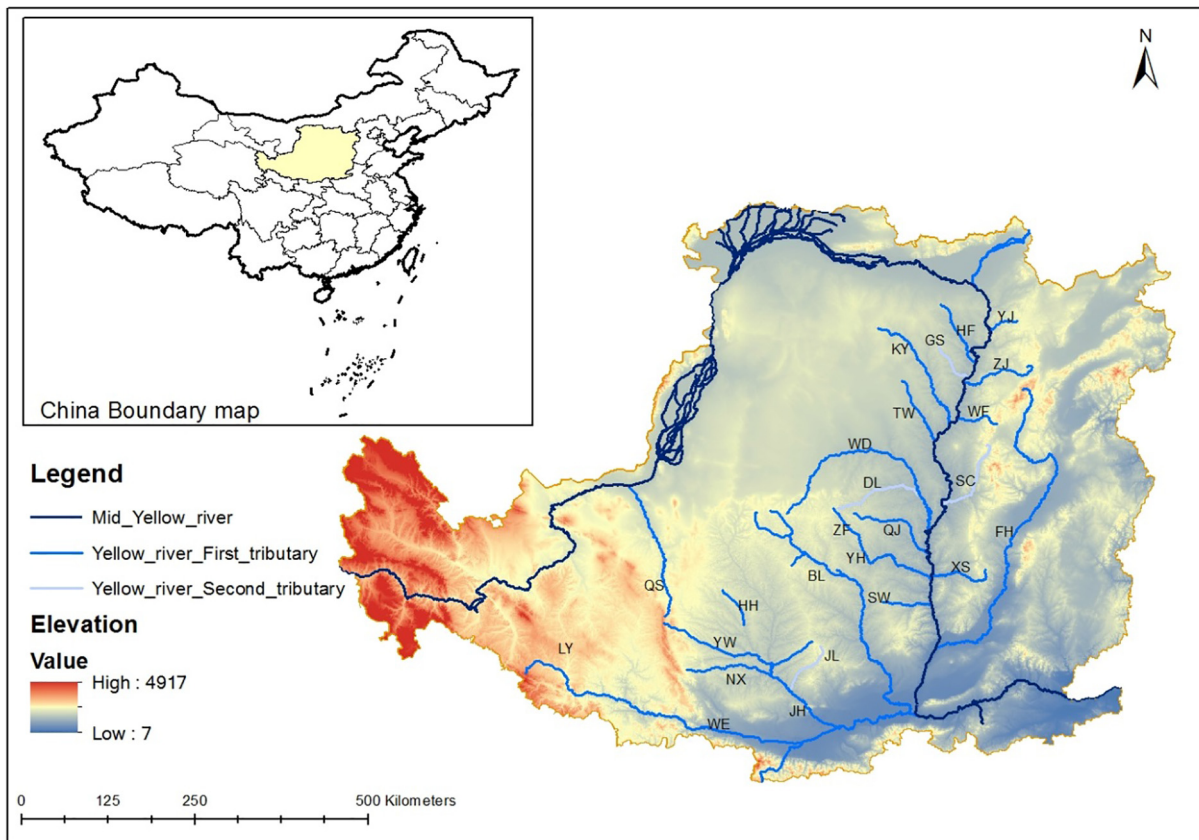


Fig. 1. Geographical location of Loess Plateau and studied rivers. Note: WE: Wei River; JH: Jing River; HF: Huangfuchuan River; WD: Wuding River; JL: Jialu River; KY: Ku Ye River; XS: Xinshui River; YH: Yan River; BL: Beiluo River; DL: Dali River; GS: Gushan River; LY: Luoyugou watershed; NX: Nanxiaohu River; QJ: Qingjian River; TW: TuWei River; YJ: Yangjiaping watershed; FH: Fen River; HH: Heihe River; YW: Yanwachuan watershed; QS: Qingshui River; SC: Sanchuan River; SW: Shiwang River; WF: Weifen River; ZF: Zhifanggou watershed; ZJ: Zhuji River.

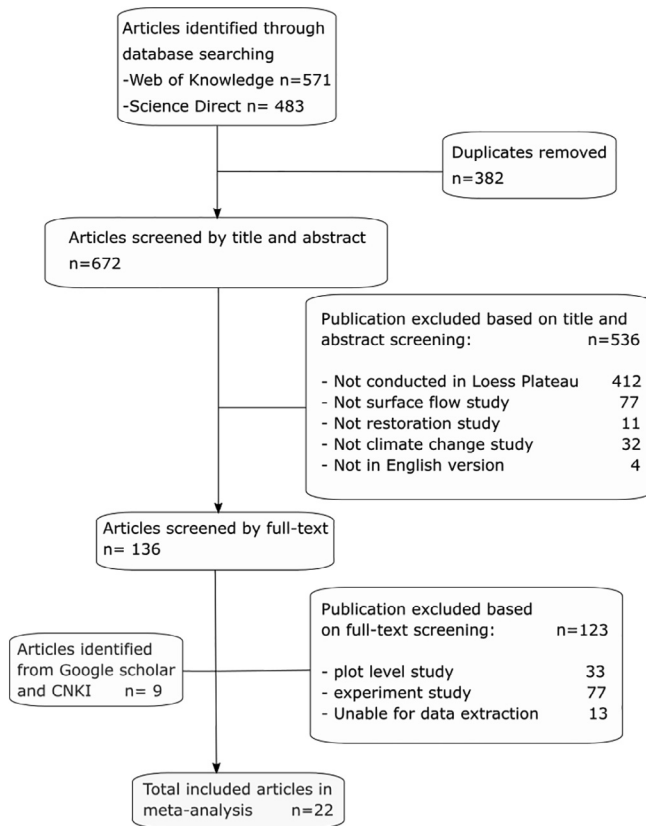


Fig. 2. Article selection flow chart.

methods of impact factors differed between articles. Double mass curves (Franke 1968) were found to be the most commonly used model (8 of 22) while the Budyko framework (REF?) and Soil and Water Assessment Tool (SWAT) model (REF?) were commonly utilized as well.

2.3. Data description

First, the characteristics of the 22 selected articles (Table. S1 Supplementary) were determined. A summary of the data obtained is given in Table 1 and shows that the 22 articles collected data covering 25 different watersheds in the Loess Plateau. The publication year of the selected articles is displayed in Fig. 3; the majority of the articles were published in 2016. Overall, this study reviewed 61 case studies from 25 watersheds from the Chinese Loess Plateau. The watershed names and frequency of the studied watersheds can be found in Fig. 4. This study conducted trend analyses based on 52 case studies and impact analyses based on 61 case studies.

2.4. Statistical methods

The main purpose was to determine the temporal trend of streamflow and climate, as well as climate and human activity impacts on the streamflow emerging from all studies. Specific indicators were used to define climate and human activity. In order to eliminate the scale difference from various watersheds in the Loess Plateau, we transferred all collected streamflow units to annual streamflow depth, which is the annual streamflow volume divided by watershed area. To analyse climatic trends, annual precipitation was selected as the main factor, because potential evapotranspiration (PET) and temperature (T) values were missing in the majority

Table 1
Categories of collected studies.

Type of studies	Number
Literature collected	22
Watersheds studied	25
Trend analysis	57
Impact analysis	61

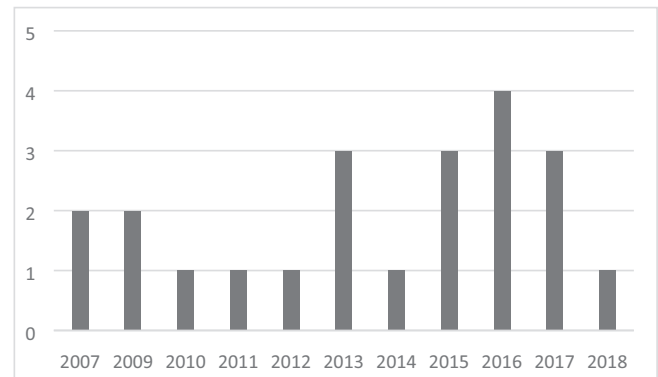


Fig. 3. The selected article frequency per publication year (n = 22).

of the articles. Additionally, the annual mean runoff coefficient (R_c) was introduced as an indicator to describe relations among precipitation and streamflow. R_c is widely used as a diagnostic variable to describe the runoff generation ability of a catchment (Merz et al. 2006), which is given by:

$$R_c = \frac{Q}{P} \quad (1)$$

where Q is the annual streamflow depth (mm) and P is the annual precipitation depth (mm). Human activity was defined as land use change and management measures, i.e. area covered by SWC measures. In this study, we grouped SWC measures into four categories: afforestation, grass planting, terrace building and dam building.

2.4.1. Mann-Kendall test

The Mann-Kendall (M-K) test (Mann 1945; Kendall 1975) is a rank-based nonparametric method used to assess the randomness of a time-series trend (Rahman et al. 2017; Xia et al. 2017). It has been widely adopted to analyze the significance of trends in hydro-meteorological data such as water quantity, streamflow, temperature and precipitation (Yue et al., 2002). In this study, the M-K test was applied to assess the trends of precipitation, runoff and runoff coefficient (R_c) in individual studies. The M-K test statistic (S) is displayed as:

$$S = \sum_{k=1}^{n-1} \sum_{j=k+1}^n \text{sgn}(x_j - x_k) \quad (2)$$

where n represents the length of the data series, x_j and x_k are sequential data values, sgn determines the sign of the corresponding elements. Under the null hypothesis the data are identically distributed with no tendency, mean μ of distribution S is expected to be 0, and the significance level σ^2 is calculated as:

$$\sigma^2 = \left\{ n(n-1)(2n+5) - \sum_{j=1}^p t_j(t_j-1)(2t_j+5) \right\} / 18 \quad (3)$$

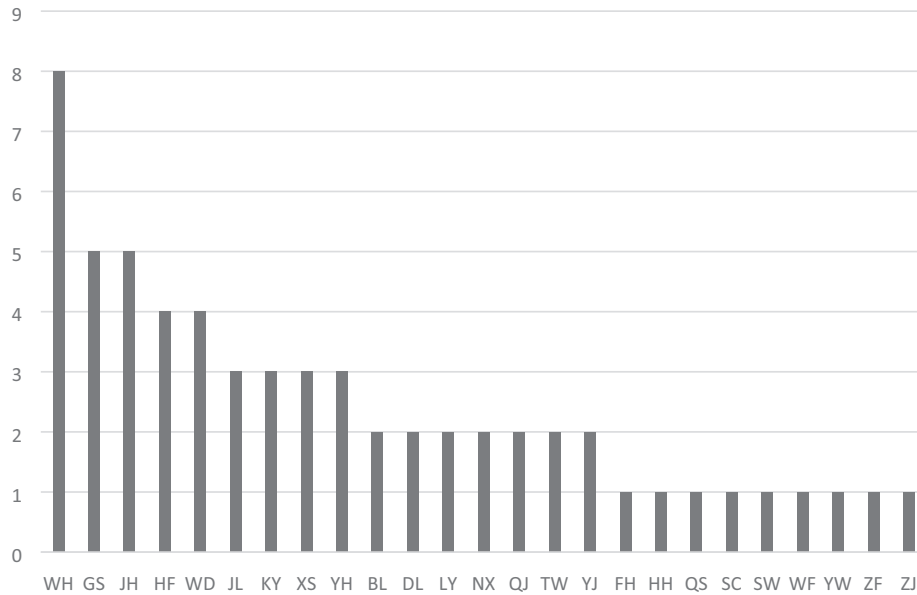


Fig. 4. Watershed studied frequency (n total = 61).

The value of Z is a standard normal distribution indicating the trend of a time series data; a positive value of Z signifies a rising trend whereas a negative value indicates a decreasing trend (Zhang et al., 2008). The Z -test statistic is calculated as:

$$Z = \text{sgn}(S)(|S| - 1)/\sigma \quad (4)$$

2.4.2. Pettitt's test

Pettitt's test is a nonparametric rank based test developed by Pettitt (1979), which is used to detect the change point in time series data. In this study, Pettitt's test was performed on annual R_c , annual precipitation and streamflow time series to determine in which year the trend started to alter. The change point calculation statistics is given by:

$$U_{t,N} = U_{t-1,N} + \sum_{j=1}^N \text{sgn}(x_t - x_j), t = 2, \dots, N \quad (5)$$

The change point T should fit in the formula $K_{t,n} = \text{Max}_{1 \leq t \leq N} |U_{t,N}|$, and the significance level of Pettitt's test is calculated as:

$$p = \exp \left[\frac{6K_{t,n}^2}{(n^3 + n^2)} \right] \quad (6)$$

2.4.3. Separating land use and climate change impacts

In the collected articles, land use change and climate change impacts were considered as the only two factors affecting streamflow, and these two impacts were calculated by various hydrological models. Generally, human activity and climate change impacts are simulated as two separated impact indices from the model, with unit percentages used to attribute effects to either driver. However, impact indices were not documented in some of the selected articles. In this case, we applied the approach from Huang and Zhang (2004) to estimate the missing impact indices. This method assumes that runoff change is caused only by variations in land management and climate, given by:

$$\Delta Q = \Delta Q_{\text{climate}} + \Delta Q_{\text{land use}} \quad (7)$$

Where:

$$\Delta Q = Q_2 - Q_1 \quad (8)$$

Calculation of the streamflow change ΔQ is based on comparison of two time periods: (a) pre-treatment period: considered as reference period assuming the initial land use; and (b) post-treatment period, after land use change occurred. Q_1 is the mean annual streamflow of the pre-treatment period and Q_2 is the mean annual streamflow of the post-treatment period. First, it is assumed that climate impacts remain the same in both pre-treatment and post-treatment period, i.e. the ΔQ is only caused by land use change in two periods (Zhang et al., 2014). A linear regression R_1 was applied between annual precipitation and annual streamflow in the pre-treatment period. By fitting annual precipitation from the post-treatment period into R_1 , the hypothetical streamflow Q'_2 without land use change impacts could be calculated. Then the separated impacts are calculated as:

$$\Delta Q_{\text{climate}} = Q_1 - Q'_2 \quad (9)$$

$$\Delta Q_{\text{land use}} = Q'_2 - Q_2 \quad (10)$$

2.4.4. Meta-analysis

In meta-analysis, the impact from an intervention is represented as effect size. Generally, impact size is used to indicate the differences between treatment and control group in experimental research, but it is also possible to identify relationships between two variables (Borenstein et al., 2011). In our study, we used R_c , precipitation and streamflow change rate as effect size which were calculated from the M-K test to compare the streamflow change level across case studies in different periods of time. Impact factors of climate change and land use change, whether directly obtained from previous studies or calculated according to the method of Huang and Zhang (2014), were calculated independently, and not related to the trend analysis. In the selected articles, methods to determine the land use and climate impacts on streamflow were not identical. Methodological differences may lead to heterogeneity blurring the true effect size. One way to treat this variability is to model it as purely random. A random effect model of meta-analysis was applied by:

$$\theta_i = \mu + u_i \quad (11)$$

where $u_i \sim N(0, \tau^2)$. The true effect size was assumed to be normally distributed with mean μ and variance τ^2 . If $\tau^2 = 0$, then $\mu = \theta_i$. μ was the change rate estimated from the M-K test, τ^2 was implied by Borenstein et al. (2011):

$$\tau = \frac{S_{diff}^2}{n} \quad (12)$$

where S_{diff} is the standard deviation of within group differences and n is the sample size. In the random effect model, the overall mean effect size among all studies is based on study weight. Under the random effect model, the weight of each study is:

$$w_i = \frac{1}{V_{yi}^*} \quad (13)$$

where V_{yi}^* is the sum of within study and between study variance.

In this study, both MK-test and Pettitt's test were performed using the R package 'trend' (<https://cran.r-project.org/web/packages/trend/index.html>). Results of the M-K test were then processed for the meta-analysis by a random effect model, which was performed using the R package 'metafor' (<https://cran.r-project.org/web/packages/metafor/index.html>) (Viechtbauer, 2010). Significance and 95% lower and upper limits for the summary effect of the meta-analysis were also calculated by the metafor package. Moreover, a meta-regression was conducted to estimate the relationship between SWC area and streamflow change. In order to understand to what extent SWC measures played a role, a meta-regression was performed between annual streamflow change rate and SWC area change rate. Streamflow change rate was obtained from the M-K test, and SWC area change rate was calculated by SWC change area divided by period of study time. The SWC indicators were defined as afforestation, grass planting, terrace building and dam building. Additionally, a categorical principal components analysis (non-linear PCA) was performed by SPSS 23.0 to determine the impact level for the different SWC measures.

3. Results

3.1. Trend of precipitation and streamflow

To determine the trend of Rc, precipitation and streamflow in recent decades, we performed a M-K test with annual time series data from the collected articles. Figs. 5 and 6 present an overview of the studies including watershed name, research duration and trend results of streamflow and runoff coefficient respectively. Overall, the majority (41 of 52) of the studied watersheds show a decreasing streamflow in recent decades and nearly half of them are significant, and none of the increasing trends was found to be significant. The highest decreasing trend is in the Wei River, with a rate of -4.91 mm/year between 1980 and 2008. Some M-K results from different articles indicated similar results, for example, in Nanxiaohe watershed, results of streamflow decrease from both Huaxing Bi et al. (2009) and Lu Xia et al. (2017) were the same (-0.47 mm/year with significance $p < 0.01$), although the research period was slightly different. The M-K results for precipitation (Fig. S2 Supplementary) show a large fluctuation among all the watersheds, ranging from -32.5 mm/year to 13.64 mm/year, however, the overall trend was found to be much less significant. M-K results for Rc were similar to those of the streamflow; 37 out of 52 case studies show a decreasing trend of Rc and none of the 12 increasing trends was found to be significant. Thus, from the M-K test results, we found a significant decrease of streamflow and Rc whereas change of precipitation was not clear. This indicates that, over the course of recent decades, less streamflow was generated from the watersheds in the Loess Pla-

teau while precipitation did not show a clear increasing or decreasing trend.

3.2. Change point in the trend

Change point detection indicates a certain time (year) that change occurred in a time series. By means of Pettitt's test, change points were determined. Fig. 7 shows the results from Pettitt's test for streamflow (Q); only the studies with a significant ($p < 0.1$) change point year ($n = 23$ out of 52) are displayed in the figure. This figure illustrates that although the research period of different case studies varied, streamflow changes occurred mainly between 1990 and 2000, concentrating at the end of the 1990 s. For the Nanxiaohe watershed the change point years were different when calculated using data from two articles (Xia et al 2017; Bi et al. 2009). We observed that a longer research period led to a later change point year, and this finding indicates that streamflow changed more sharply after 2008 in the Nanxiaohe watershed.

In contrast, and in line with the findings for the trends given in the previous section, very few (2 of 52) significant change points were found in precipitation time series (Table. S3 Supplementary), which indicates no sharp precipitation changes happened in the studied watersheds. Similar to the results from streamflow (Fig. 7), change points of Rc showed 18 significant values in total, and the majority of these points concentrate between 1990 and 2000, with 9 of the changes occurring in 1999 and early 2000 (Fig. 8). For Nanxiaohe watershed, we observed a large variation in change point across different articles; this may due to precipitation variance.

3.3. Meta-analysis

In order to summarize a general trend of streamflow among all selected watershed studies, a random effect model was constructed. Fig. 9 shows how streamflow change rates of different watersheds are distributed in the meta-analysis study. Overall, the majority of the streamflow studies show a decreasing trend. Meanwhile the red point indicates the mean effect size of the analysis, illustrating a significant ($p < 0.01$) decreasing rate of -0.46 mm/year as extracted from the 52 case studies. Thus, from Fig. 9 we can deduce that in the past decades, the majority of the watersheds in the Loess Plateau experienced a reduction of flow. Compared with the streamflow results, precipitation (Fig. S6 Supplementary) showed a smaller within study variance, and the mean effect size displayed an insignificant trend with value -0.53 mm/year. This indicates that precipitation did not have a certain trend in the studied watersheds. Fig. 10 is quite revealing in several ways. First, the Z value is relatively concentrated on the negative side, for instance, in Jinghe watershed, a -5.52 Z value was detected from (Zhang et al. 2016) with 2.33% contribution to the mean effect size. The overall negatives indicate that Rc is decreasing continuously. In general, Fig. 10 presents a significant decreasing Z value of Rc with -1.40 /year. We can deduct from these results that the majority of the watersheds in the Loess Plateau were capturing, storing and transpiring more precipitation and therefore less runoff was generated.

3.4. Land use change impacts

Land use change impacts on streamflow were obtained from two sources: a) directly collected from selected articles; b) estimated using the approach of Huang and Zhang (2004) from streamflow and precipitation time series data. Total land use change and climate change impact index values collected are displayed in Fig. 11. This figure illustrates that overall land cover

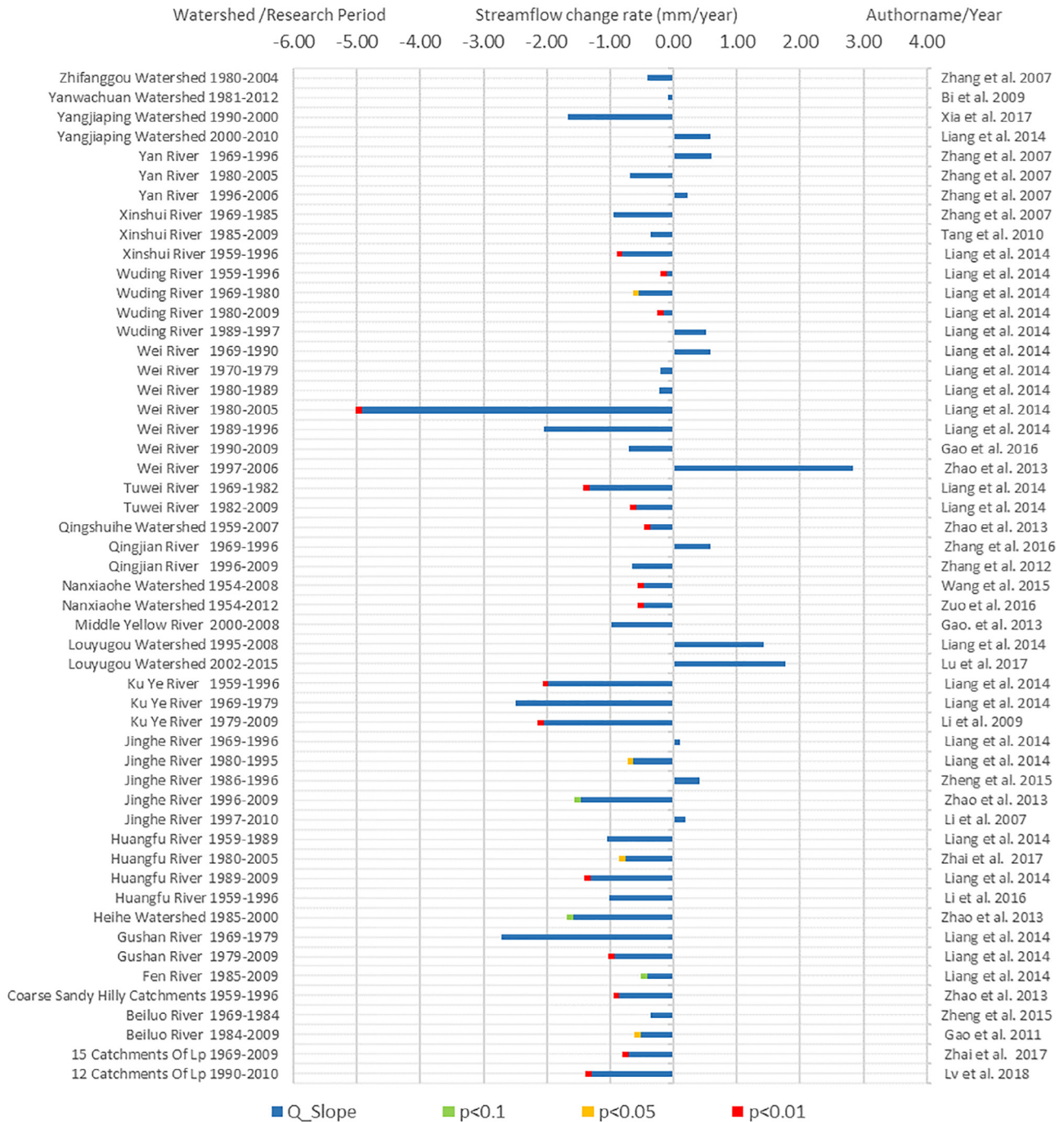


Fig. 5. Streamflow change rate of 52 watershed case studies. Note: colors on top of the bars indicate the significance level.

change more strongly influenced streamflow change compared to climate change on the Loess Plateau. On the basis of the results from M-K and Pettitt's test, 1999 seems to be a change point of streamflow quantity on the Loess Plateau. Thus, we categorized the impact index according to the time period of study years, followed by conducting a meta-analysis based on the land use impact index. A climate change impact meta-analysis was not conducted, because it is assumed the sum of land use and climate change impact index is 100%. It is apparent from Fig. 12 that compared to studies after 1999, land use impact index of earlier studies have a wider between-study variance. Wider between-study variance illustrates that there is a larger difference in land use impact index between different watersheds. The impact of land use on stream-

flow increased from 55% in the period before 1999 to 68% in the period after 1999, with a weight of 57.60%. This sharp increase may be due to the GTGP's initiation and the dramatic land cover change it brought. Overall, the mean effect size of a total of 61 case studies indicates that land use change impacts account for 63.52% of the total streamflow change whereas climate change accounts for the rest (i.e. 36.48%).

3.5. Impact of soil and water conservation measures

Fig. 13 shows the meta-regression results for streamflow change rate and SWC separately (a–d), as well as for the total net SWC area (e). Blue circles in the figure represent watershed case

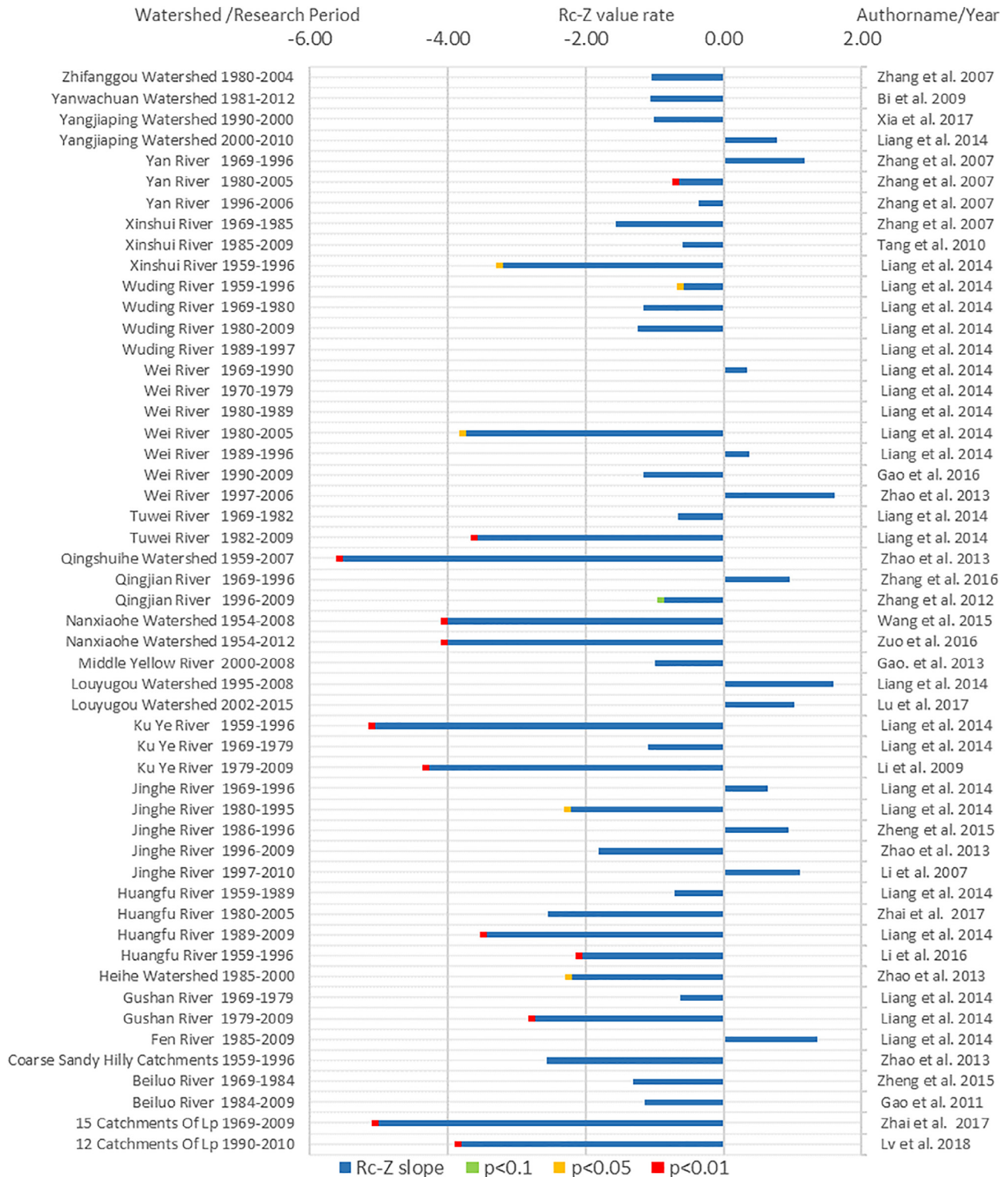


Fig. 6. Runoff coefficient change of 52 watershed case studies. Note: colors on top of the bars indicate the significance level.

studies, whereby the diameter of the blue circle is inversely related with within study variance of streamflow change. For example, in Fig. 13-b, the two big blue dots in the middle indicate two studies with smaller variance compared to other studies. The solid red line is the regression line between the change rate in SWC area and change in water flow quantity, the dotted red lines represent the confidence interval of the meta-regression. Generally, there is no

strong correlation between the four different SWC measures and the streamflow change rate. Afforestation (a) and terracing (d) show a slight stimulation effect on streamflow reduction whereas grass planting (b) and dam building (c) are restraining the streamflow. Besides, compared to afforestation and grass planting, in the majority of the watersheds, terrace and dam area increased very slowly and concentrate on the left edge of the graphs, close to

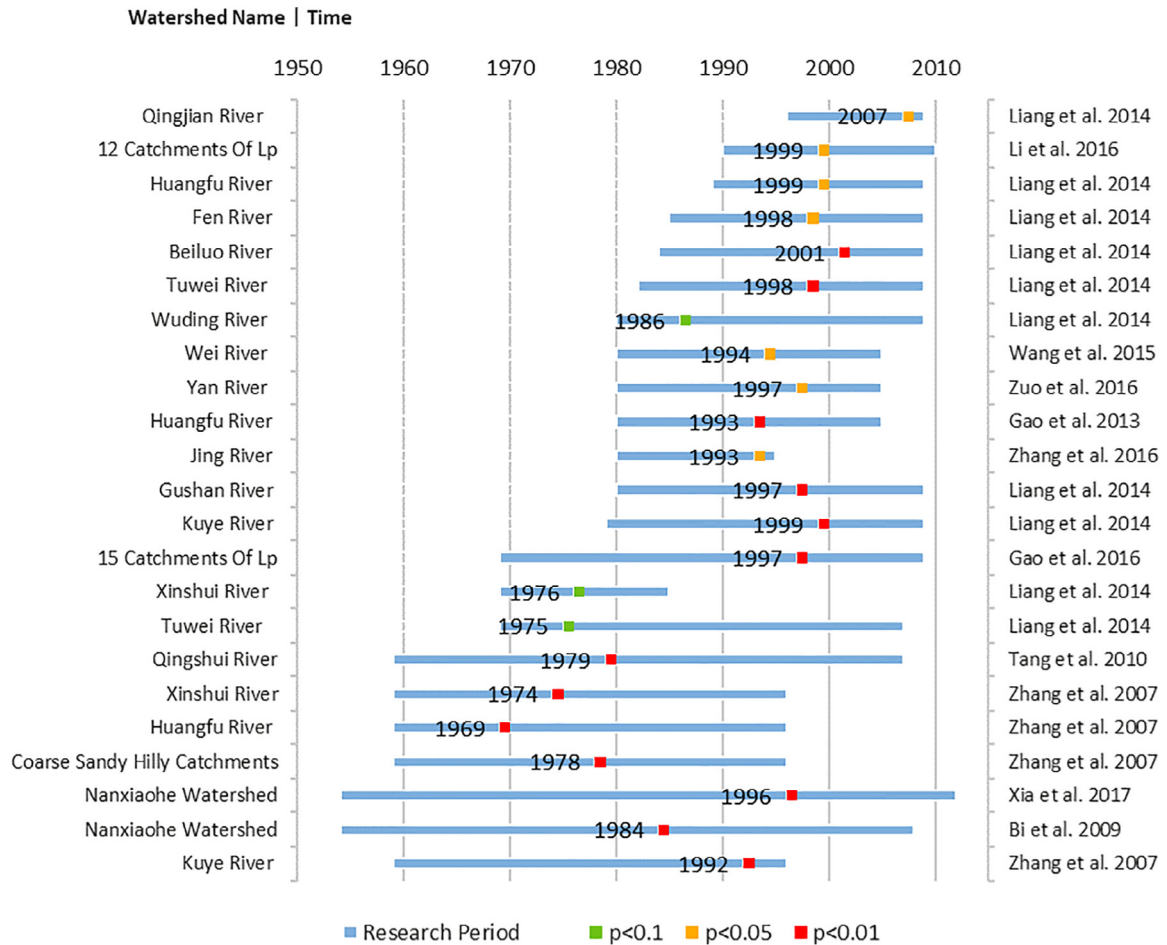


Fig. 7. Change point of streamflow in watershed case studies. Note: cubes in the figure indicate the significance level ($n = 23$).

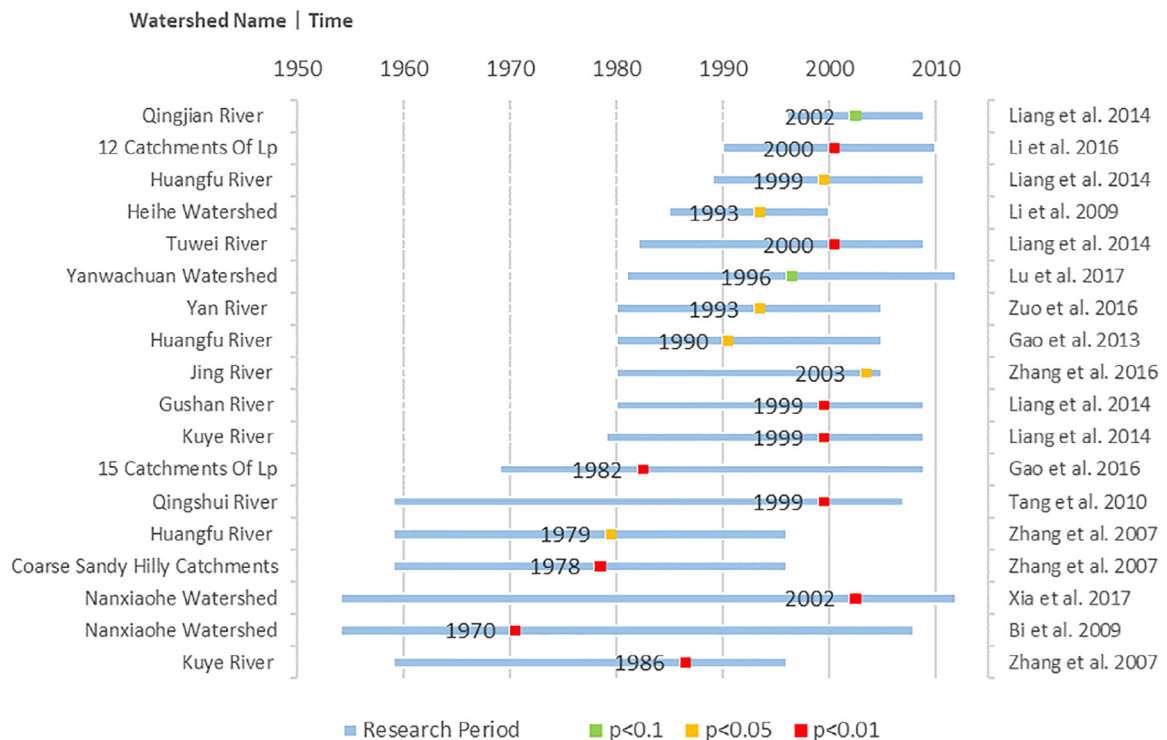


Fig. 8. Change point of runoff coefficient in watershed case studies. Note: cubes in the figure indicate the significance level ($n = 18$).

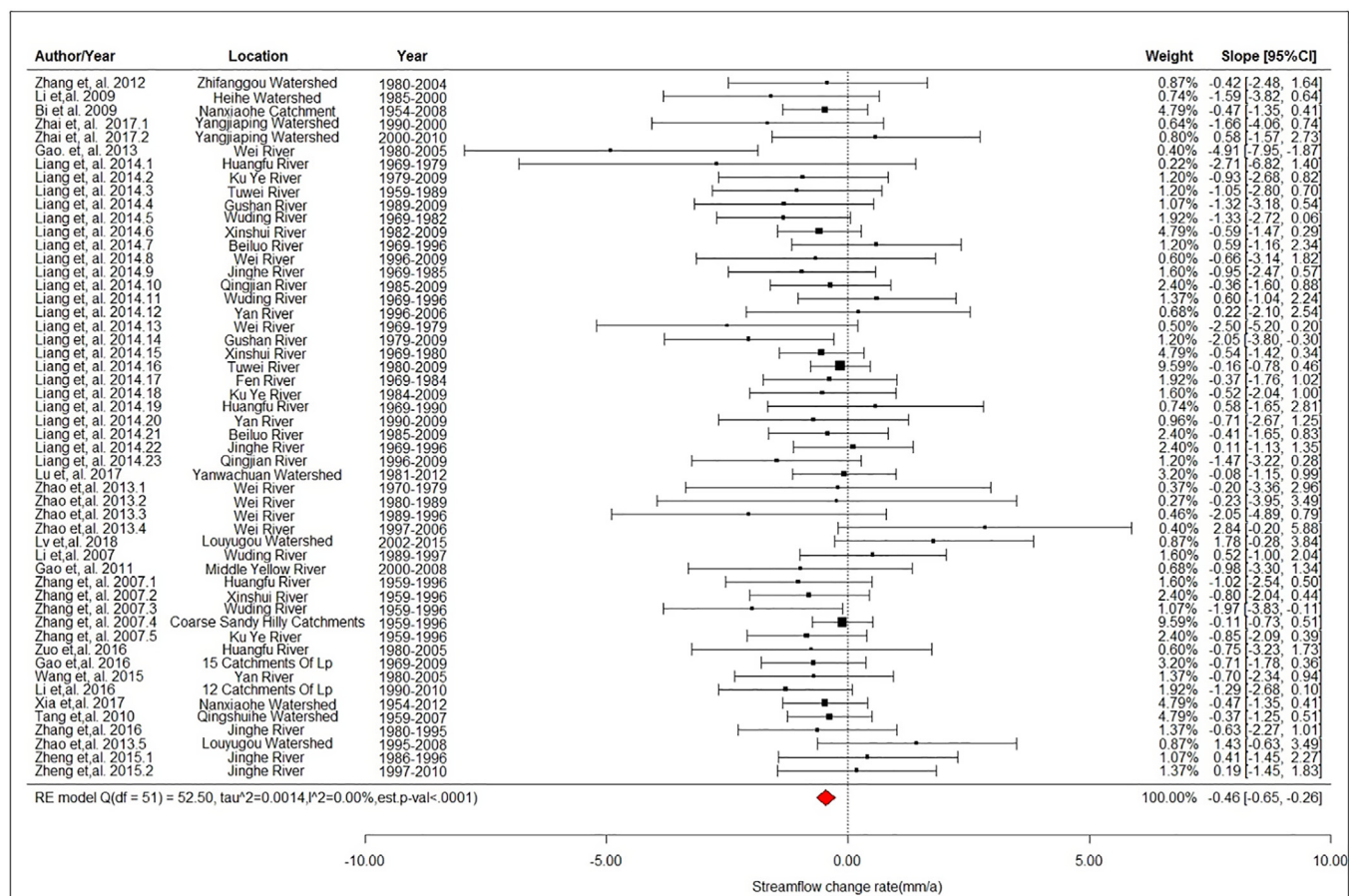


Fig. 9. Meta-analysis result of streamflow change rate.

0%/year. However, when considering the total net SWC area change rate (e), overall, we observe that the increase in the SWC area causes a decrease of streamflow.

Fig. 14 shows the results of a categorical principal components analysis (PCA) performed to determine the relationships between SWC measures and streamflow change. The proportion of variance accounted for in the first and second component is 61.68% and 32.83% respectively, meaning that the sum of the component's variance accounts for a considerable proportion (94.50%). From the PCA plot we observe that principle component (PC) 1 was affected by terrace building and afforestation while PC 2 was strongly influenced by streamflow change. In line with the results from the meta-regression, streamflow change and SWC change area displayed an angle close to 90°, which indicates a low correlation between them. However, Fig. 13 illustrates that the level of influence of the SWC area on streamflow change was following the order: total restoration area > terrace building > afforestation > grass planting > dam building.

4. Discussion

Many recent studies show evidence that many rivers on the Loess Plateau are experiencing a streamflow reduction in recent decades (Tian et al. 2019; Wu et al. 2017; Miao et al. 2011). We have found similar results in this study. By using a meta-analysis we calculated a $-0.46 \text{ mm year}^{-1}$ streamflow reduction across the selected watersheds over different time scales ranging from 1959 till 2015. Based on the selected hydrological modelling studies, we found that land use change caused 63.25% of the stream-

flow decrease, while Hu et al. (2017) found that ecological restoration accounts for 72.18% of annual runoff reduction based on a meta-analysis of field experimental studies. The difference may be caused by indicator diversity, scale variety and methodological differences. In this study, total land use change on a watershed scale was considered in the hydrological models, whereas field experiments only considered ecological restoration impacts on a plot scale. Based on our results, the precipitation change trend was found to be insignificant. At the same time, it was found based on analysis of the reported importance of land use versus climate change that the latter still accounted for 36.48% of streamflow change. Selection of the indicators may cause this difference, as potential evapotranspiration (PET) and temperature data were not included in the trend analysis due to insufficient data from literature. Individual studies have found different impacts of climate change. For example, Zhao et al. (2014) found that climate change is causing 87% of flow reduction in Beiluo river by using precipitation, PET and temperature as indicators, while in our study the precipitation change is found insignificant in Beiluo river in the same period.

Meta-analysis provides a new method to summarize a general meteorological trend on regional scale, for example, Jiang et al. (2017) found that annual precipitation decreased by -1.52 mm/year from 1959 to 1999 and increased from 2000 to 2012 by 3.47 mm/year on the entire Loess Plateau, while our meta-analysis produced a different result, i.e. no significant precipitation change trend was found. The results of Jiang et al. (2017) were based on the weighted value of 49 weather stations from the whole Loess Plateau whereas our results were estimated from each watershed study. The Loess Plateau covers an area of

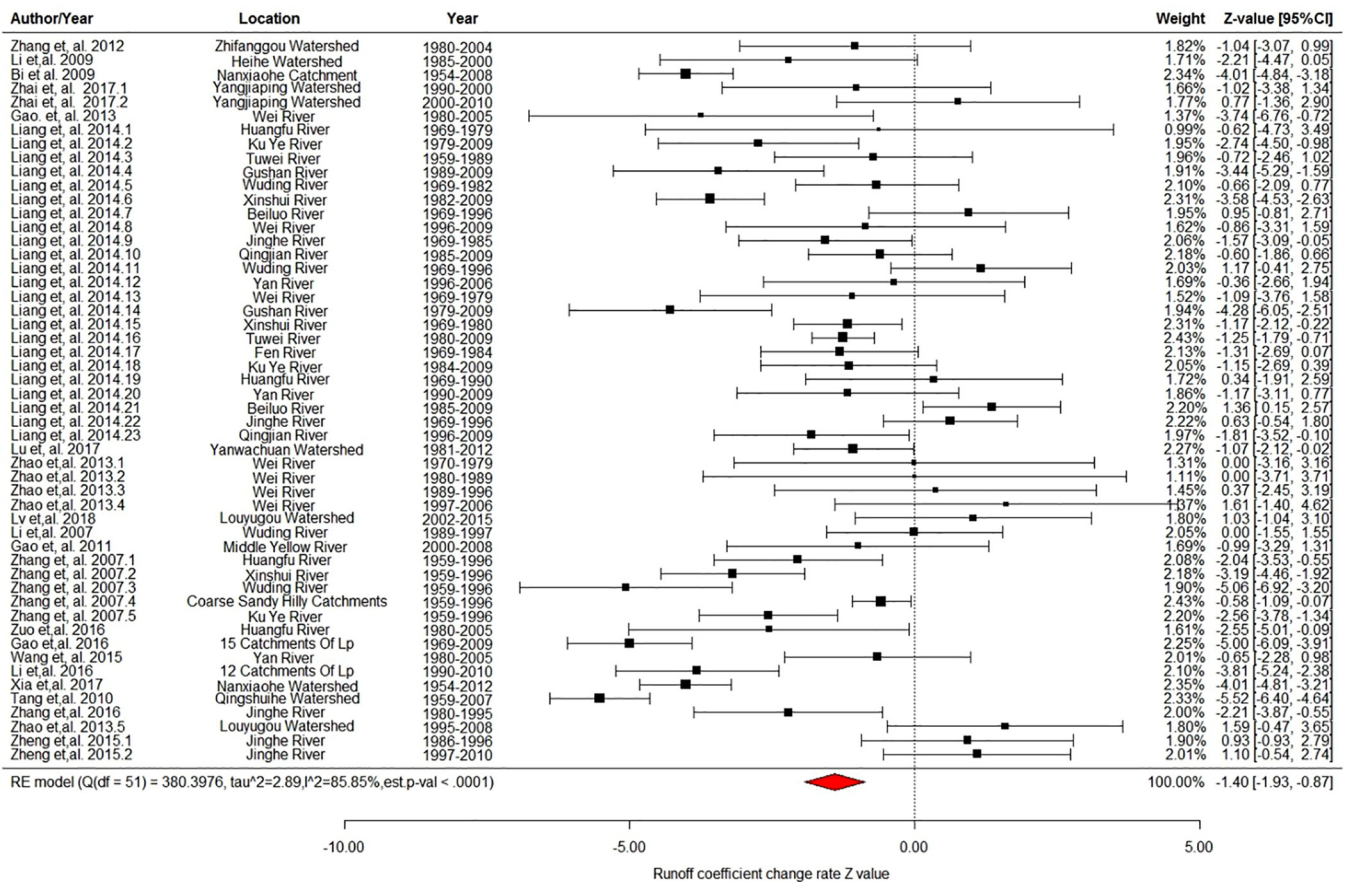


Fig. 10. Meta-analysis result of runoff coefficient change rate.

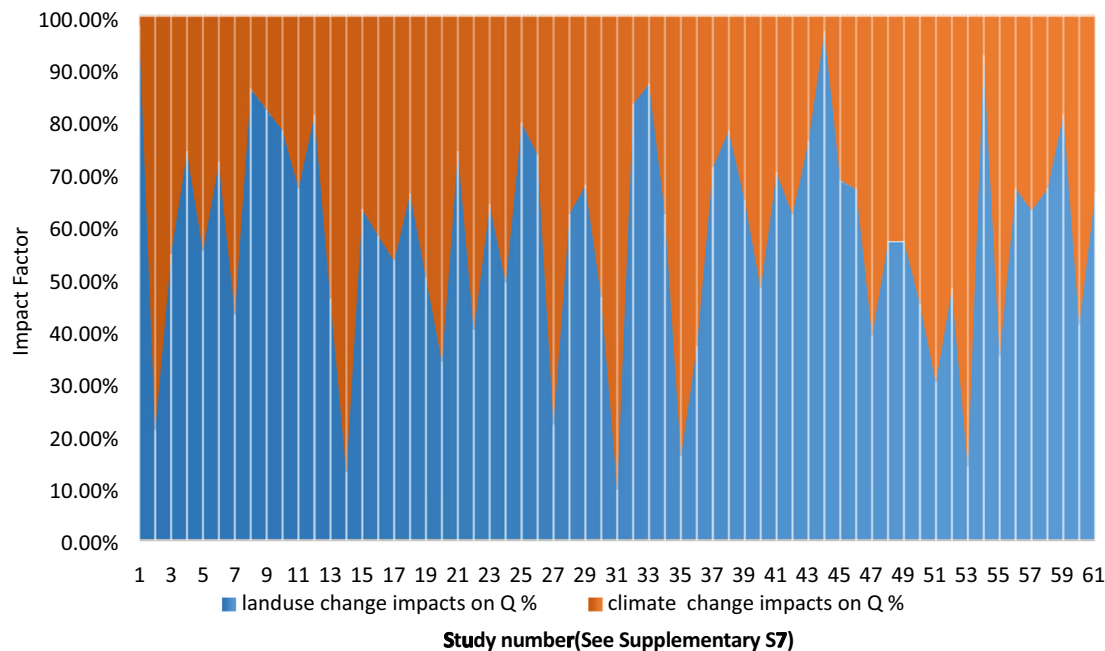


Fig. 11. Land use change and climate change impacts on streamflow change (n = 61) Note: Watershed name, author and research period are listed in Fig. S7 Supplementary.

640,000 km² of land and precipitation varies from region to region (Ma et al. 2019); therefore, we think it is more representative to derive the precipitation trend for this regional scale from multiple

watershed studies instead of using the mean value from 49 distant stations. The main objective of this study was to describe the streamflow dynamics on the Loess Plateau based on different

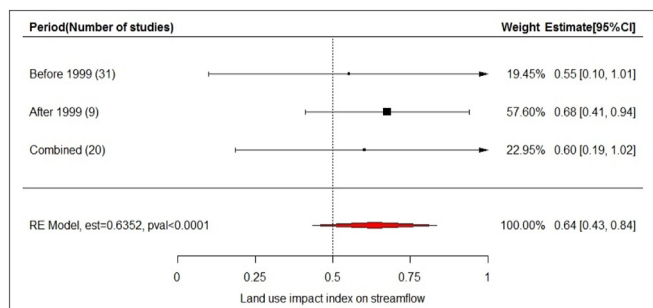


Fig. 12. Land use impact factor on streamflow change in the Loess Plateau in different periods.

watershed-level analyses in previous studies. By looking at results in the same watershed from different publications at similar periods of time, we recognize differences among the results. For example, results from [Gao et al. \(2013\)](#) showed a decrease in streamflow of -4.91 mm/year in the Wei River between 1980 and 2005 whereas results from [Zhao et al. \(2013\)](#) showed an increase of 2.84 mm/year in the Wei River between 1997 and 2006. Thus, it is necessary to include all the watershed studies from various research periods by different authors to describe a more comprehensive overview of the streamflow change in the Loess Plateau.

Climate change is regarded as one of the key factors affecting streamflow, which consists of precipitation change, temperature change, PET change, wind speed change and humidity change ([Wu et al. 2017](#); [de Hipt et al. 2019b](#)). In the majority of the previous studies on the Chinese Loess Plateau, precipitation was consid-

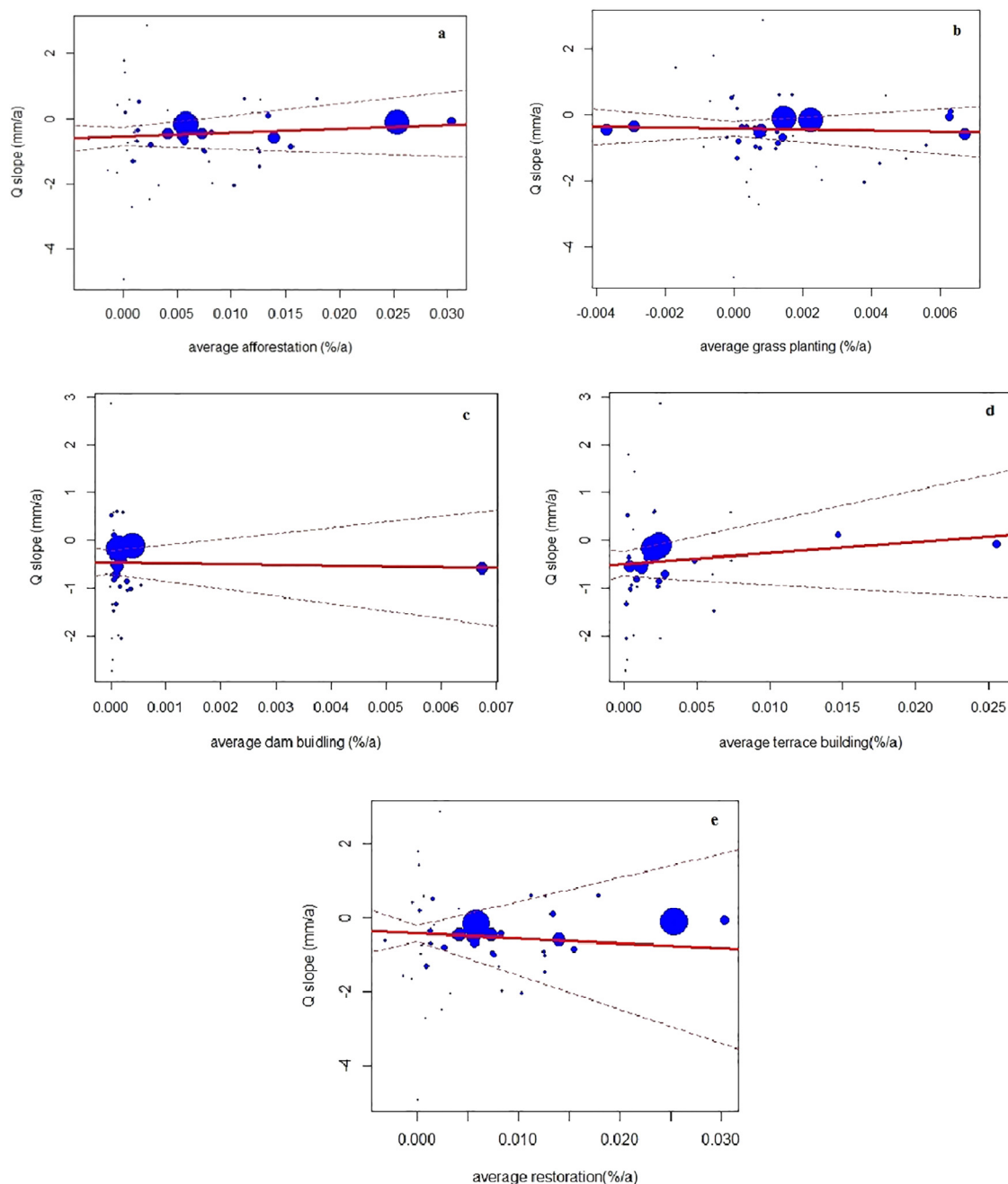


Fig. 13. Meta-regression between SWC area change rate and streamflow slope.

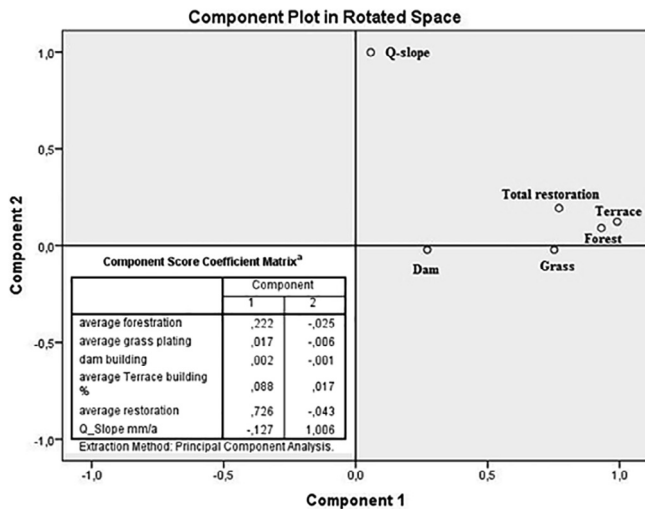


Fig. 14. Principle component analysis of the dataset on streamflow change and soil and water conservation measures.

ered as the primary indicator to understand climatic impacts on streamflow ((Lv et al. 2018; Xu 2011; Zhao et al. 2016). Tang et al. 2013 showed that precipitation contributes the most to runoff change among net radiation, air temperature, wind speed and relative humidity in the Chinese Yellow River basin. In our study, precipitation was selected as an indicator to represent climate change in the Chinese Loess Plateau, and the results showed that there was no significant precipitation change among 52 case studies. Additionally, we conducted a trend and meta-analysis for PET and T data extracted from the 22 articles. However, only 5 and 3 articles had good data on temperature and PET, respectively. The results can be found in [supplementary material](#) (Table. S5 and Fig. S6); both T and PET showed an increasing trend while only the increase of PET is significant according to the meta-analysis. However, this is based on very limited data.

In a meta-analysis, since published studies are more likely to be considered than unpublished data, any bias in the literature is possibly reflected in the meta-analysis as well; this issue is generally known as publication bias (Borenstein et al. 2011). Non-significant studies have been estimated 61–86% less likely to be published than significant studies; therefore results of these non-significant studies may be ignored by the literature reviewers (Chan et al. 2004). Additionally it is likely that some studies were missed using our search criteria. In order to identify the publica-

tion bias in this study, funnel plots (Fig. 15) were created. Funnel plots display the relation between variance and effect size. In a funnel plot, studies are expected to show symmetry at the top, few studies missing in the middle, and more studies missing near the bottom (Borenstein et al. 2011). From the funnel plot of streamflow (Fig. 15; left) we can deduce some studies with lower significance of streamflow change are likely to be missing, whereas in the precipitation funnel plot (Fig. 15, right) some studies with higher significance of precipitation change are missing on the top. Overall, the funnel plots indicate the meta-analysis of streamflow change is more reliable than the results of precipitation due to less publication bias. To some extent, this deduction could explain why the climate index accounts for 36.48% of the streamflow reduction while few significant trends were found in the precipitation change.

In this study, a significant reduction of streamflow is observed while an expansion of forest and grassland occurred at the same time. Similar results were found by Lü et al (2012): they found that runoff decreased by 10.3 mm/year whereas forest and grassland area increased by 4.9% and 6.6% respectively on the whole Loess Plateau in 2002–2008. GTGP is found to be the primary driving factor for this vegetation increase (Feng et al. 2013). One argument to explain this trade-off between water yield reduction and SWC area increase is the increased water consumption from land restoration. A higher evapotranspiration rate from the restored vegetation is found to be a crucial reason for runoff-coefficient reduction (Li et al. 2016). Meanwhile deep soil moisture decreased more than 35% after land use conversion and a soil moisture deficit appeared in all types of newly restored land (Yang et al. 2012; Su and Shanguan 2019). According to the findings from this and previous studies, a negative effect of land restoration is exposed, and a long-term monitoring process is required to assess the degree of streamflow reduction.

Starting from 2010, there has been a tremendous increase in meta-analysis studies; however, the quality of the meta-analyses varies (Gurevitch et al. 2018). In this study, we conducted a meta-analysis with regular procedures including literature selection, set up of effect size, meta-regression and publication bias analyses, which is considered to be a complete meta-analysis (Borenstein et al. 2011). Further refinements of this study could be achieved; for instance, this study processed the statistics based on annual data, while seasonal and event data could be used to investigate the impacts of precipitation on the streamflow more precisely, e.g. for low flow and flood conditions (Gai et al. 2019). In some selected watersheds, hydrological models were used to

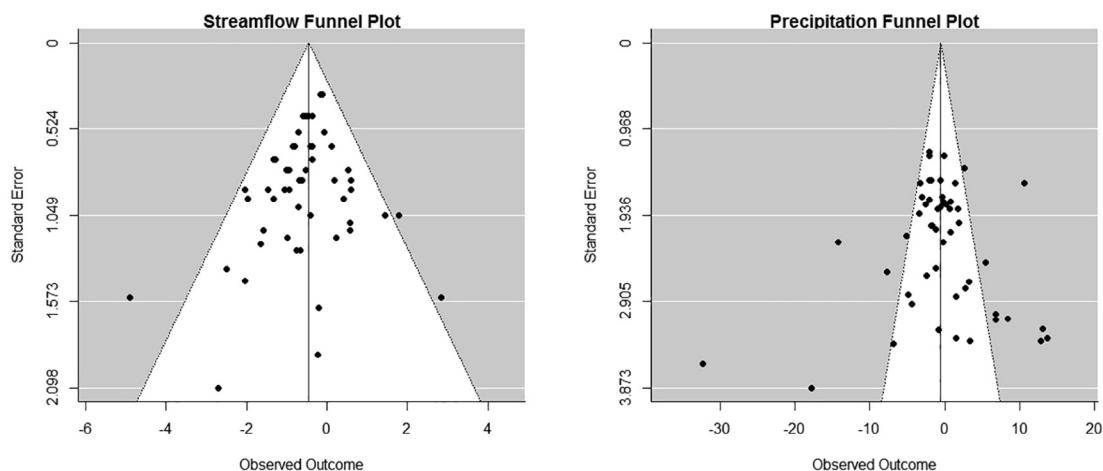


Fig. 15. Funnel plot from meta-analysis of streamflow change rate and precipitation change rate.

simulate scenarios in different land cover and climate conditions; a meta-analysis can compare the results from different scenarios on the condition that data is available and complete.

5. Conclusions

This is the first study that estimates streamflow dynamics from a multitude of watersheds on the entire Loess Plateau. Meta-analysis was used to minimize the diversity of results caused by differences in methods in the selected studies. The streamflow on the Loess Plateau shows a significant decreasing trend of -0.46 mm/year over the period 1959–2015, while no significant change was found in precipitation and temperature. From the meta-analysis, we identified that land cover change accounts for 63.52% of the streamflow change whereas climate variability accounts for 36.48%. There was a small negative correlation between streamflow change and net change of SWC area. The level of influence of different SWC measures on streamflow reduction is in the order of: restoration area > terrace building > afforestation > grass planting > dam building. The results indicate that the continuously implemented land restoration actions on the Loess Plateau by the Chinese government are one of the major drivers of streamflow reduction in recent decades. It is recommended for policy makers to consider the spatial-temporal trends in water demand of restored and revegetated land, and to keep climate change impacts in mind while designing future land restoration in order to avoid its side-effects on the hydro-environment.

CRedit authorship contribution statement

Hao Chen: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Resources, Software, Project administration, Validation, Visualization, Writing - review & editing, Writing - review & editing. **Luuk Fleskens:** Methodology, Supervision, Conceptualization, Writing - review & editing. **Jantiene Baartman:** Methodology, Supervision, Writing - review & editing. **Fei Wang:** Funding acquisition, Supervision. **Simon Moolenaar:** Funding acquisition, Supervision, Writing - review & editing. **Coen Ritsema:** Funding acquisition, Supervision.

Declaration of Competing Interest

The authors declare that there is no conflict of interest.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.scitotenv.2019.134989>.

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